

RUSK COUNTY AP, WI, USA

WMO: 723758

Lat: 45.497N

Lon: 91.001W

Elev: 1238

StdP: 14.05

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.7	-10.2	-22.7	1.6	-13.5	-17.6	2.2	-8.7	22.6	13.2	20.0	16.2	3.8	300	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	86.1	71.8	83.1	69.9	81.0	67.9	74.5	82.3	72.7	79.9	70.9	77.5	9.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	125.7	78.5	70.3	116.8	76.6	68.6	109.9	75.0	38.9	82.4	37.1	80.2	35.5	77.8	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.7	15.9	-23.9	91.0	3.9	3.1	-26.7	93.2	-29.0	95.0	-31.1	96.8	-34.0	99.1		
(5)			-24.2	77.9	3.8	2.3	-27.0	79.6	-29.2	80.9	-31.3	82.2	-34.1	83.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	14.6	16.1	30.8	43.1	55.8	64.5	69.0	66.5	59.5	46.6	33.4	20.1		
	DBStd	21.49	13.25	12.44	12.51	9.58	8.62	6.11	5.50	5.30	8.09	8.77	9.94	11.49		
	HDD50	4559	1097	950	608	246	37	1	0	0	14	174	504	928		
	HDD65	8230	1562	1370	1061	658	309	80	25	46	203	574	949	1393		
	CDD50	2173	0	0	12	38	215	437	590	510	298	68	5	0		
	CDD65	372	0	0	1	0	22	66	150	91	38	4	0	0		
	CDH74	3200	0	0	2	11	249	588	1331	711	284	24	0	0		
	CDH80	698	0	0	0	0	57	113	343	131	53	1	0	0		
	WSAvg	6.6	6.8	7.0	7.0	8.1	7.2	6.2	5.4	5.1	6.0	6.9	7.2	6.7		
	PrecAvg	32.8	1.1	0.9	1.8	2.9	3.5	4.7	3.9	4.3	3.8	2.8	1.9	1.2		
(15) Precipitation	PrecMax	43.1	3.6	2.7	4.0	6.4	6.5	7.9	7.5	10.2	8.2	6.1	7.0	2.0		
	PrecMin	25.2	0.2	0.1	0.5	0.6	1.6	2.1	1.6	0.7	0.8	0.9	0.0	0.3		
	PrecStd	5.1	0.7	0.6	0.9	1.4	1.5	1.7	1.6	2.0	1.8	1.5	1.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	49.2	70.3	75.2	86.4	87.2	90.9	87.5	85.8	77.4	64.1	46.1		
		MCWB	36.9	43.6	59.5	57.8	69.1	73.2	76.0	72.6	71.9	64.4	53.6	41.4		
	2%	DB	37.1	40.8	60.9	69.8	80.5	83.0	86.4	83.5	81.1	72.0	58.2	40.7		
		MCWB	35.2	36.8	52.8	54.4	64.6	68.9	72.4	71.3	68.9	62.4	50.9	38.6		
	5%	DB	34.8	36.4	53.8	64.3	76.3	80.3	83.7	80.9	77.2	67.2	52.2	36.7		
		MCWB	33.5	33.2	46.5	51.3	62.3	66.6	70.8	68.2	66.5	58.4	46.7	34.9		
	10%	DB	32.5	32.7	46.8	59.9	71.8	77.3	81.2	78.1	73.4	61.2	47.5	34.6		
		MCWB	31.3	30.5	41.5	48.7	60.1	64.6	68.5	67.2	64.7	54.6	43.2	33.4		
	0.4%	WB	37.6	43.4	60.4	60.5	73.4	75.8	78.8	76.2	74.2	71.0	57.1	43.7		
		MCDB	39.5	48.0	68.0	68.3	83.6	83.6	87.0	83.2	82.2	74.5	61.0	44.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.5	37.2	54.8	56.7	68.5	72.0	74.9	73.5	71.5	63.2	51.5	38.3		
		MCDB	36.9	40.0	58.4	66.9	75.1	80.6	83.3	79.7	77.2	69.5	57.1	40.2		
	5%	WB	33.9	33.5	46.7	53.3	64.9	69.2	72.8	71.7	69.2	59.4	47.0	35.3		
		MCDB	34.9	35.7	52.2	61.9	72.3	76.3	80.4	78.0	74.5	66.0	51.0	36.6		
	10%	WB	31.6	30.9	41.9	49.9	62.0	67.0	71.0	69.6	66.7	54.9	43.5	33.5		
		MCDB	32.5	32.6	46.6	59.1	68.9	74.1	78.1	75.1	72.5	60.2	47.5	34.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	19.2	18.5	19.5	21.9	20.4	20.6	20.4	20.5	17.2	14.1	13.4		
		MCDBR	14.6	18.0	23.5	28.6	26.6	23.4	23.2	23.0	23.3	24.6	22.1	13.5		
		MCWBR	12.7	14.6	14.7	15.9	13.1	11.6	11.2	11.6	12.5	14.9	15.5	11.3		
		MCDBR	13.3	15.9	20.1	25.0	21.7	19.5	20.2	19.0	19.0	20.9	20.7	12.3		
	5% WB	MCWBR	11.9	13.1	13.9	15.6	12.2	11.5	11.1	10.9	11.4	14.2	15.4	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.268	0.301	0.344	0.374	0.375	0.372	0.376	0.352	0.312	0.284	0.257		
	taud		2.391	2.334	2.387	2.396	2.346	2.363	2.395	2.396	2.453	2.550	2.527	2.441		
	Ebn at Noon		280	292	296	291	284	283	282	277	275	274	264	264		
	Edh at Noon		23	29	32	35	38	38	36	35	30	24	20	20		
(44) All-Sky Solar Radiation	RadAvg		517	870	1230	1446	1685	1856	1933	1621	1264	795	479	354		
	RadStd		56	71	114	149	127	147	82	114	81	89	49	41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page